

D1 – Early testing of the code showed examples of race conditions. Two users accessing a shared resource were able to read, write, and/or modify the same account, producing incorrect balances. The shared resource had to have an atomic lock to ensure the critical section was locked at the appropriate times.

The proof in Milestone 3 helped our group think about what belongs in the critical section. Prior to coding, we were able to examine how deposits, withdrawals, and balance inquiries functioned, which helped us understand how those individual components would interact with the inclusion of the semaphore.

Another insight we gained was how small changes in the operations could break the functionality of the program. Making sure the code reflected the milestone design was important in both theory and practice to achieve accurate results.